



Atomic and Electronic Structure of Ultrathin Iron Film on MgO(001) Surface

Toshio URANO and Toru KANAJI

Faculty of Engineering, Kobe University, Rokkodai, Nada, Kobe 657

(Received May 28, 1988)

Iron film deposited on a MgO(001) surface grows layer by layer and forms an epitaxial film at room temperature. Low-energy electron diffraction (LEED) I-V curves are measured for films of various thicknesses. In comparison with the curves by theoretical analysis it is found that an iron film grows pseudomorphic and iron atoms sit just above oxygen ions at the stage of one monolayer (1 ML) average thickness. The structure of an iron film is body-centered tetragonal (bct) in the early stage, but it begins to change to bcc structure at about 10 Å average thickness and at the same time a widening of the valence-band electronic state is observed by Auger electron spectroscopy (AES).

[iron film, MgO(001) surface, surface structure, LEED, I-V curve, AES]

§1. Introduction

From a viewpoint of the reliability and the durability of semiconductor devices, investigations for microscopic behaviour of metal-semiconductor and metal-insulator interfaces are important. For example, most transition metals react on silicon and form silicide at rather low temperatures.

On the other hand, the film growth by metal vapour deposition offers opportunities synthesizing metastable crystalline phases which have novel physical properties. For example, it is reported that fcc iron (γ -iron) epitaxial films are stable on Cu(001) with one to four monolayers.¹⁾

In our preliminary experiment it seemed that iron film deposited on MgO(001) surface grew layer by layer in the early stage and finally formed an Fe(001) epitaxial film at room temperature.²⁾ The purpose of this paper is to clarify the atomic structure of iron films at various thicknesses by means of low-energy electron diffraction (LEED) and to investigate the electronic state by Auger electron spectroscopy (AES).

§2. Experimental Results

The substrate crystal was prepared by the following method. In air, the MgO crystal

(Daiichi Kigenso Kagaku Corp., Osaka) was cleaved to a size of about 5 mm $\times$ 5 mm $\times$ 1 mm and was mounted on a sample holder made of tantalum foil. After a conventional evacuation process, usually, a remarkable Auger signal of carbon was detected. Therefore, the specimen was heated up to 800°C in oxygen atmosphere at a pressure of about 1×10^{-3} Pa until the Auger signal of carbon disappeared. The purity of the MgO crystal used for this study was not confirmed, but for another piece of the crystal provided from the same manufacturer the principal impurities were Al₂O₃ (50 ppm), CaO (30 ppm) and Fe₂O₃ (20 ppm).

Iron atoms were evaporated from an iron wire (Materials Research Corp., Marz grade) which was directly heated by an electric current. Its diameter was 0.03 inch, and its purity was 4N. In order to avoid an effect of the impurities, the iron wire was preheated for about 10 hours at the same temperature during the film formation. The temperature was measured by an optical pyrometer and the change of evaporation rate was monitored by a crystal monitor. The absolute deposition rate was determined by the following method. A small glass plate was placed during the preheating at the position where the substrate crystal was placed during the actual experi-

ment. This glass plate was taken out after the experiment and the thickness of the deposited film was measured with the interference-stripes method.

The experiments were carried out with the conventional four-grid LEED-AES system. LEED I-V curves were measured by a simple semiautomatic system which was developed by the authors based on a spot-photometer method.³⁾ The primary electron beam current was about $2\ \mu\text{A}$ and, practically, this value did not change by the change of the electron energy in the region higher than 100 eV. The change of light transparency through the grids slightly deforms the profile of the experimental I-V curve, however, it seems that this effect is not important because the position of peaks and dips should not be affected.

In Fig. 1, the changing rates of the normalized Auger peak amplitudes of Mg-LVV (26 eV), O-KLL (501 eV), Fe-MVV (42 eV) and Fe-LMV (648 eV) during the film formation at room temperature are shown. The deposition rate was about $0.2\ \text{\AA}/\text{min}$. The four curves in the figure separate from one another in accordance with individual energy (i.e., escape length). Consequently, it is recognized that an iron film grew layer by layer on the MgO(001) surface from Palmberg's criterion.⁴⁾ In spite of several repetitions of corresponding measurements, however, a distinct break point could not be observed. Therefore, it is difficult to apply the Jackson's criterion.⁵⁾

Although three Auger peaks except the Fe-MVV peak change smoothly, the Fe-MVV Auger peak indicates an unusual behaviour, namely, (i) increasing rate of normalized peak amplitude is considerably slow, (ii) once at about 55 min (corresponding to about $11\ \text{\AA}$) increasing rate has halted. The cause of this behaviour should be discussed in §4.

In the early stage, as the thickness of the film increased, the brightness of the beams of 1×1 LEED pattern decreased gradually, but did not disappear. From $10\ \text{\AA}$ thickness the brightness increased again gradually with increasing the film thickness, and finally a pattern corresponding to a bulk-like Fe(001) surface appeared. Typical LEED patterns for various film thicknesses are shown in Fig. 2.

In order to study a surface structure more

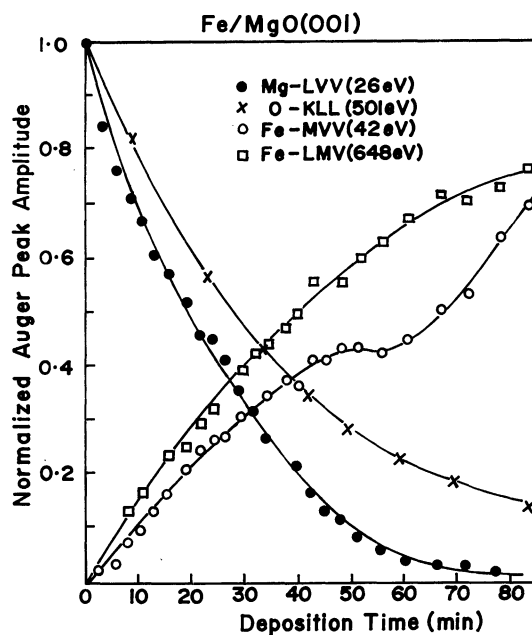


Fig. 1. Normalized Auger amplitudes vs time of deposition for iron film on MgO(001) surface at room temperature. Deposition rate is about $0.2\ \text{\AA}/\text{min}$.

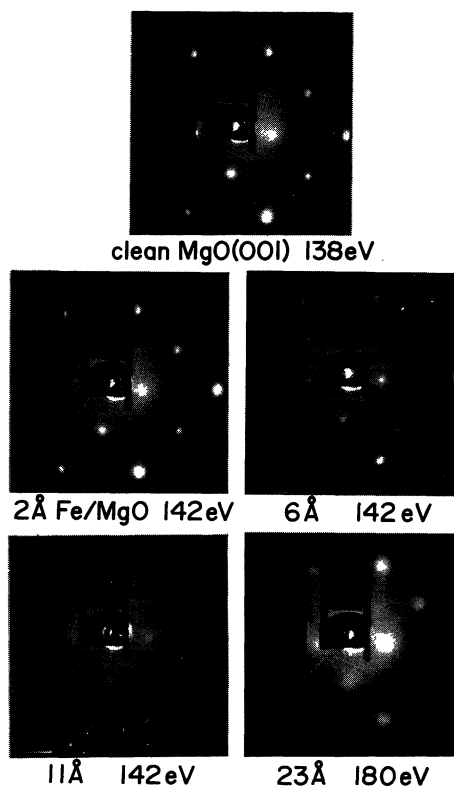


Fig. 2. Typical LEED patterns of iron films of various thicknesses on MgO(001) surface.

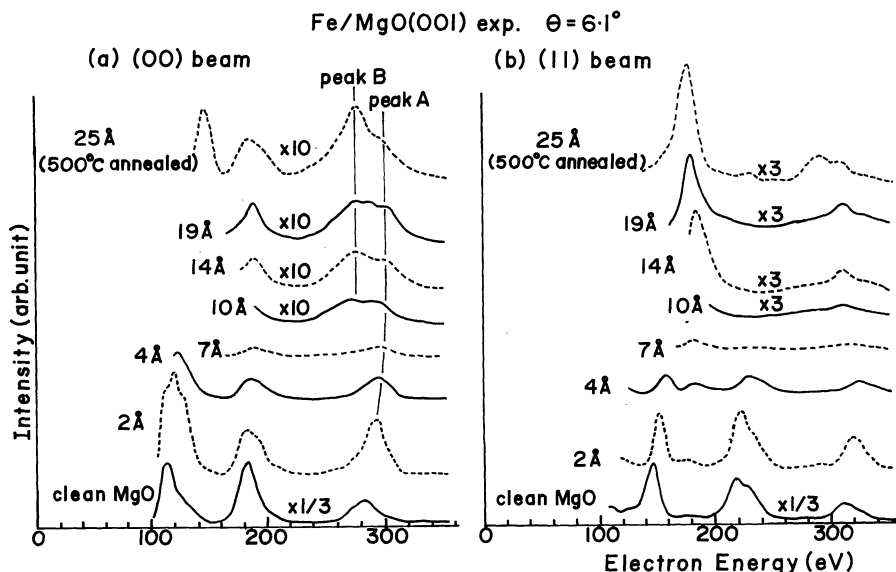


Fig. 3. Experimental I-V curves of (a) (00) and (b) (11) beams for iron films of various thicknesses on MgO(001) surface. The low energy ends are limited by the charging up phenomena.

precisely, LEED I-V curves for (00) and (11) beams were measured under the condition of incident angle $\theta=6.1^\circ$ at the stage of various film thicknesses and are shown in Fig. 3. The low-energy end of these curves is limited by the charging up phenomena. From this figure the following considerations are deduced.

(a) In the early stage it seemed that an iron film was pseudomorphic, because only a 1×1 LEED pattern was observed. For the film of 2 Å average thickness (about 1 ML), most peaks of both (00) and (11) beams shifted slightly to the higher energy side. It is likely that these shifts represent a decrease of lattice spacing between the topmost two planes.

(b) Up to about 10 Å, the crystallinity was imperfect, since the beam intensity decreased and the background intensity increased.

(c) At about 10 Å, another structure (probably bulk-like bcc iron) began to grow. This stage corresponds to the time at which unusual behaviour of the Fe-MVV Auger peak amplitude was observed. From this time the crystallinity became better again with increasing the film thickness.

§3. Comparison of Experimental I-V Curves with Results of Dynamical Calculations

In order to determine the sit site of the iron

atom on the substrate crystal and the atomic structure of the film, dynamical calculations have been carried out. The computer program by Pendry⁶⁾ was partly modified to apply to the system in which plural atoms exist in a two-dimensional unit cell and to apply to the overlayer system. The layer stacking is handled with the renormalized forward scattering (RFS) perturbation method.

The parameters used to calculate a set of phase shifts of Mg and O ions are indicated in ref. 7, that is, the core state wave functions tabulated by Herman and Skillman⁸⁾ are used, the MgO crystal is fully ionic, the muffin-tin radii for Mg and O ions are assumed to be 0.65 and 1.46 Å, respectively, and the Madelung potential is uniform within a muffin-tin sphere.

On the other hand, the electron configuration of the iron atom is assumed to be 6 3d core state electrons and 2 conduction electrons. The muffin-tin radius is one half of the nearest neighbour distance for bcc iron crystal.

Thirty-seven beams in all, that is, (00) beam and sets of {10}, {11}, {20}, {21}, {22}, {30} and {31} beams are included in the calculation. The number of angular momentum included is 5, that is, $l_{\max}=4$ and the temperature

is assumed to be 300 K. What effect each parameter has on a result of calculation is considered in ref. 7 and a sufficient number of parameters which do not have a bad influence upon a result are included.

First, in order to determine the sit site of iron atoms, experimental I-V curve for the film of 2 Å is compared with theoretical curves for a pseudomorphic film of 1 ML thickness. It is indicated in ref. 7 that for the MgO(001) clean surface the theoretical I-V curve with no relaxation or with the smallest one (2.5% expansion) fits best with the experimental curve. Further, it is indicated that the degree of the surface rumpling is small and the change of it due to heat treatment cannot be recognized. Then, the MgO(001) substrate is regarded as a truncated surface of a bulk crystal and is assumed that this structure does not change due to the deposition of iron atoms.

Three adsorbed sites, that is, just above the oxygen ion, above the magnesium ion and the

bridge site, are considered. In Fig. 4, calculated I-V curves of the (00) beam for various spacings ($d_{\perp}$) between an iron plane and a topmost MgO(001) plane for three adsorbed sites are shown together with the experimental I-V curve for the film of 2 Å. The energy value of calculated I-V curves is shifted by the inner potential of $V_{or}=15$ eV. From this figure a possibility for the adsorbed site of just above magnesium ion is excluded. Though the agreement is not sufficiently excellent to conclude quantitatively, three possible positions, that is, just above the oxygen ion at about 2.0 Å height and two bridge sites at about 2.0 Å and 1.3 Å height, are considered.

Next, the experimental I-V curve of (11) beam is compared with theoretical curves for the plausible structures and is shown in Fig. 5. From this figure, the site of just above the oxygen ion at 2.0 Å height is most probable. Although the distance between the oxygen and

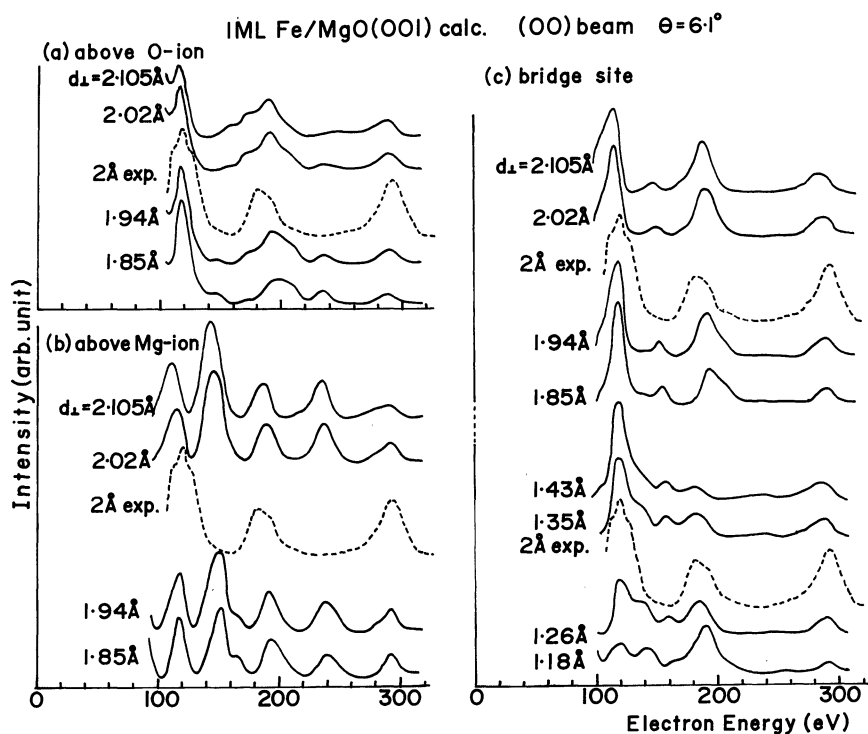


Fig. 4. Calculated I-V curves of (00) beam for the monolayer pseudomorphic iron film on MgO(001) surface with different spacing ($d_{\perp}$) between the iron plane and the topmost MgO(001) plane (solid curve) for comparison with the experimental curve of 2 Å average thickness (broken curve). Three sit-sites of iron atom are considered, (a) above O-ion, (b) above Mg-ion and (c) bridge site.

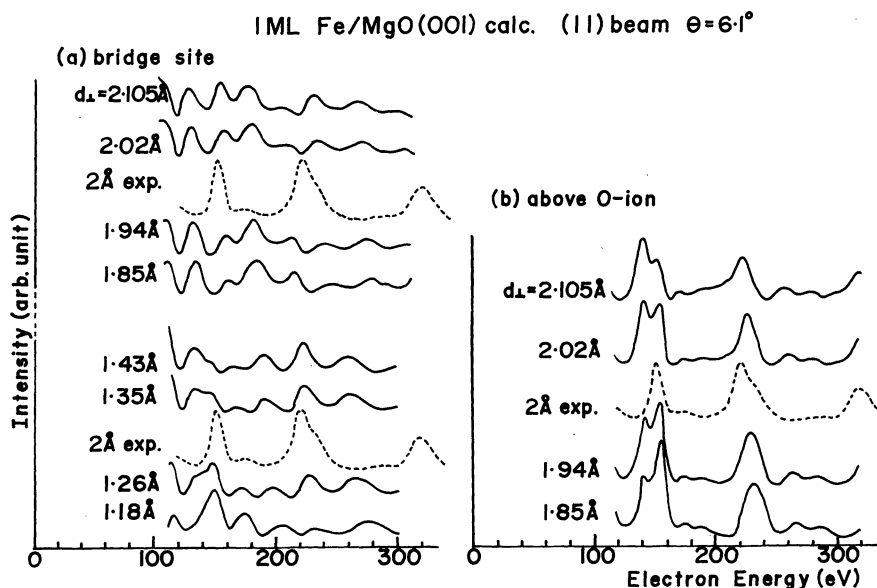


Fig. 5. Calculated I-V curves of (11) beam for the monolayer iron film on MgO(001) surface (solid curve) for comparison with the experimental curve (broken curve), (a) bridge site and (b) above O-ion.

iron atoms is slightly shorter than the corresponding values of 2.14 Å in FeO (NaCl type) and 2.10 Å (Fe I) in Fe₃O₄ (spinel),⁹⁾ this value is within the limits of permission. The disappearance of the fine profile which is probably based on a dynamical effect in the experimental I-V curve seems to be caused by the lack of crystal perfection.

Second, the I-V curve of (00) beam for bulk Fe(001) surface was calculated and was compared with the experimental curve obtained by Horiguchi and Nakanishi¹⁰⁾ for the clean Fe(001) surface. Two I-V curves were very similar to each other with respect to the peak and dip positions except for a few insignificant differences of peak shape. Thus, it is ascertained that the calculated curve has a sufficient reliability.

The same theoretical curve is compared with the top curve in Fig. 3(a) of the (00) beam (500°C annealed after 25 Å deposition) and these curves are shown in Figs. 6(b) and 6(a). Two principal peak positions at about 145 eV and 275 eV coincide very well. Because the peak at 275 eV corresponds to the peak B in Fig. 3(a), it is considered that a growth of bcc iron film begins at an average thickness of 10 Å.

The origin of peak A in Fig. 3(a) is considered as follows. The lattice constant of a two-dimensional square lattice for MgO(001) surface is larger by 4% than that for the Fe(001) surface. In the early stage, because an iron film grows in pseudomorph, it is con-

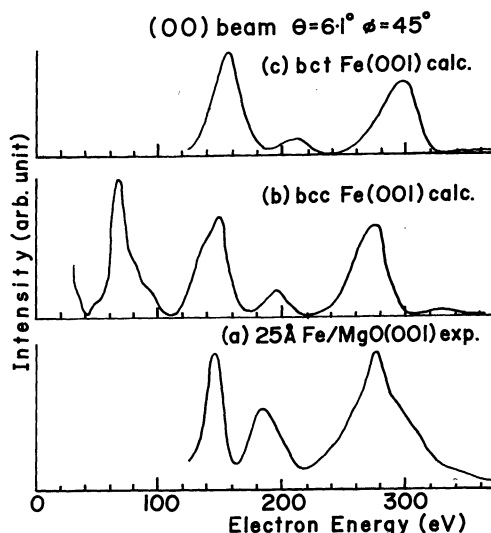


Fig. 6. (a) Experimental I-V curve of (00) beam for iron film of 25 Å average thickness after annealing at 500°C, and calculated I-V curves of (00) beam for Fe(001) surface of (b) bcc and (c) body-centered tetragonal (bct) structures.

sidered that the lattice spacing between the topmost two iron planes perpendicular to the surface has shortened. Two cases have been considered about the degree of shortening. One is the case of which the nearest neighbour distance between iron atoms is equal to that of bulk bcc iron crystal ($a_{nn}=2.48 \text{ \AA}$). In this case the lattice spacing $d_{\perp}$ perpendicular to the surface becomes $1.31 \text{ \AA}$ (namely, for bcc iron crystal $d_{\perp}=1.43 \text{ \AA}$). I-V curves were calculated for this structure, however, the peak positions did not coincide.

Another is the case of which the value is estimated from the elastic constant¹¹⁾ and the lattice spacing becomes $1.375 \text{ \AA}$. The calculated I-V curve of (00) beam for this body-centered tetragonal (bct) structure is indicated in Fig. 6(c). The energy position of the peak at about 300 eV is consistent with that of the peak A in the experimental curve in Fig. 3(a). Therefore, it is considered that the iron film with an average thickness less than $10 \text{ \AA}$ has a bct structure. For the (11) beam an analogous consideration mentioned above can be applied.

According to this model, the iron film of less than $10 \text{ \AA}$ average thickness has a bct structure, and in the iron film of more than $10 \text{ \AA}$ thickness, domains of bct Fe and bcc Fe structures coexist and the domain of bcc Fe increases with increasing average film thickness.

§4. Electronic Structure of Iron Films

It is indicated in Fig. 1 that the Fe-MVV Auger peak (peak-to-peak amplitude in $dN(E)/dE$ curve) shows a curious behaviour at 55 min (about $11 \text{ \AA}$ average thickness). Three reasons can be considered as a cause of this behaviour, (a) the sticking coefficient of coming flying atoms has changed, (b) the film growth mode has changed, or (c) an electronic state which is concerned with the MVV Auger transition has changed. Of these, in the case of suspected causes of (a) and (b) other Auger signals also must show the same abnormal behaviour. It becomes clear in §3 that the growth of the bcc structure has begun at the same time as this curious behaviour. It is not surprising that the electronic state has changed in accordance with the change of the surface structure. Then, the suspected cause (c) seems to be most probable.

Usually, due to the restriction of signal-to-noise (S/N) ratio it is regarded that the peak-to-peak amplitude in the derivative curve ($dN(E)/dE$ curve) of an energy distribution curve ($N(E)$ curve) is regarded as an Auger peak intensity. This is reasonable as far as the peak shape in the $N(E)$ curve is analogous, that is, the peak width does not change. In the case in which the peak shape changes, however, the peak-to-peak amplitude does not show the intensity of the Auger peak correctly and an integrated intensity of the Auger peak in $N(E)$ curve must be estimated. Therefore, the integrated intensity of the Fe-MVV Auger peak has been estimated.

Since it is difficult to measure the peak intensity in the $N(E)$ curve directly due to the restriction of S/N ratio, it has been obtained by integrating the $dN(E)/dE$ curve. Unfortunately, the Fe-MVV Auger peak (42 eV) exists at the position where the slope of background due to the true secondary electrons is very steep, then, this background has been subtracted from a raw $dN(E)/dE$ curve approximately, at first. Next, this curve has been integrated.

At various stages of average film thickness, this procedure has been carried out repeatedly and the peak width and the integrated intensity of the Fe-MVV Auger peak in $N(E)$ curve have been estimated. In Fig. 7, from the top, (a) peak-to-peak amplitude in $dN(E)/dE$ curve, (b) peak width in $N(E)$ curve and (c) an integrated intensity in $N(E)$ curve calculated manually are shown, respectively. At the stage of unusual behaviour in the curve (a), the peak width becomes wide in the curve (b). Consequently, though the curve (c) is not thoroughly smooth, a curious behaviour is rather moderated.

It is considered that the width of Fe-MVV Auger peak has changed because the valence-band electronic state is very sensitive to the atomic structure, particularly, to the nearest neighbour distance. It is reasonable to consider that the electronic state has changed simultaneously when the structure of the film has changed. In §3, it is suggested that the iron film has begun to change from body-centered tetragonal (bct) structure to bcc structure at $10 \text{ \AA}$ average thickness. In this case, the

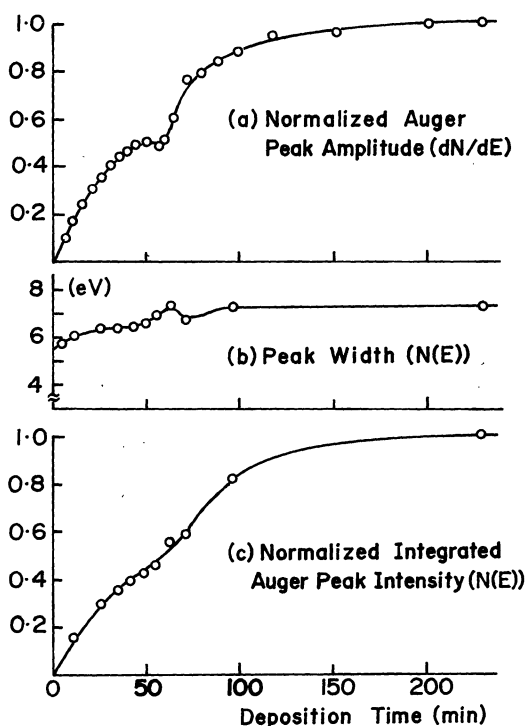


Fig. 7. Change of the Fe-MVV Auger signal: (a) peak-to-peak amplitude in the $dN(E)/dE$ curve, and (b) peak width and (c) integrated intensity in the $N(E)$ curve.

nearest neighbour atomic distance changes from 2.51 Å to 2.48 Å.

Heine has indicated that the d-bandwidth at Γ point shrinks in proportion to the inverse fifth power of the lattice constant.¹²⁾ According to this concept, because the nearest neighbour atomic distance in bcc phase is smaller than that in bct phase by 1.2%, the bandwidth becomes broad by about 6%.

In the MVV Auger transition, because the valence-band electronic state concerns twice, the energy distribution curve of this Auger peak reflects self-convolution of the density of state. Then, roughly speaking, the Auger peak width in $N(E)$ curve in bcc phase is broader than that in bct phase by about 12%. In Fig. 7, it is shown that at the stage of 55 min the peak width has changed from 6.3 eV to 7.2 eV. This change corresponds to the roughly estimated value.

§5. Summary

Iron film deposited on MgO(001) surface

grows layer by layer and forms a bcc Fe(001) epitaxial film finally at room temperature. By the comparison of experimental LEED I-V curves with theoretical curves, it is considered that at the stage of 1 ML thickness the iron film is pseudomorphic and the iron atom sits just above an oxygen ion. Moreover, iron film begins to change from body-centered tetragonal structure to bcc structure at about 10 Å and at the same time a widening of valence-band electronic state has been observed by AES.

It is not clear that why the increasing ratio of the normalized Fe-MVV Auger peak amplitude is slow, however, it can be considered that a part of the charge of iron atoms may transfer to the substrate. In order to get more precise knowledge, angle-resolved ultraviolet photoelectron spectroscopy (ARUPS) measurements are in preparation, because photoemission can be regarded as a one-electron process and it should give more simple evidence.

Acknowledgements

The authors wish to express thanks to Professor N. Kunitomi of the Osaka University (at present, Hiroshima Institute of Technology) for his helpful discussions and encouragement. The authors would like to thank Dr. M. Kaburagi of Kobe University for helpful discussions. The authors are pleased to acknowledge the cooperation of Mr. Y. Taki on the part of this study. This work is partially supported by the Grant-in-Aid for Scientific Research on Priority Areas by the Ministry of Education, Science and Culture.

References

- 1) M. Onellion, M. A. Thompson, J. L. Erskine, C. B. Duke and A. Paton: *Surf. Sci.* **179** (1987) 219.
- 2) T. Kanaji, T. Kagotani and S. Nagata: *Thin Solid Films* **32** (1976) 217.
- 3) T. Kanaji, H. Nakatsuka, T. Urano and Y. Taki: *Surf. Sci.* **86** (1979) 587.
- 4) P. W. Parmberg and T. N. Rhodin: *J. Appl. Phys.* **39** (1968) 2425.
- 5) D. C. Jackson, T. E. Gallon and A. Chambers: *Surf. Sci.* **36** (1973) 381.
- 6) J. B. Pendry: *Low Energy Electron Diffraction* (Academic press, London, 1974).
- 7) T. Urano, T. Kanaji and M. Kaburagi: *Surf. Sci.* **134** (1983) 109.

- 8) F. Herman and S. Skillman: *Atomic Structure Calculations* (Prentice Hall, Englewood Cliffs, NJ, 1963).
 - 9) W. B. Peason: *Handbook of Lattice Spacing and Structures of Metals and Alloys* (Pergamon press, London, 1958).
 - 10) T. Horiguchi and S. Nakanishi: *Proc. 2nd Int. Conf. Solid Surfaces, Kyoto, 1974*, Jpn. J. Appl. Phys **13** (1974) Suppl. 2, Pt. 2, p. 89.
 - 11) C. Kittel: *Introduction to Solid State Physics* (Wiley, New York, 1956) 2nd ed., Chap. 4, p. 93.
 - 12) V. Heine: Phys. Rev. **153** (1967) 673.
-